From Detection to Attribution: Tracking and Explaining Stream Change in Antarctica from Terrain Alone

Summary. A tool exists that finds where Antarctic streams are reshaping the land, but it stops at location: it says where, never why. This project starts at that line. The work will connect the measured change to the processes driving it (heat, melt, thawing ground), make the detector work across different sensors and all four valleys, and attach a real uncertainty value to every pixel.

1. The setup and the gap

The McMurdo Dry Valleys (MDVs) are the largest ice-free region in Antarctica, a cold polar desert of bare rock and gravel. For roughly a century they were treated as the most geomorphically stable place on Earth, but that's no longer holding. The system sits right at the energy threshold for melting, so small increases in absorbed heat are enough to melt ice and degrade permafrost. The visible result is ephemeral streams: channels that flow for a few weeks each summer, transport sediment, and reshape the channel before going dry again.

Those streams are the clearest climate signal available in an otherwise stable landscape. If they accelerate, that's an early indicator worth taking seriously.

Barlow (2026) [1] built the tool that measures them, and it rests on three components worth naming:

1. Terrain-only segmentation. A U-Net (an image-segmentation CNN) traces stream channels from the shape of the ground alone (elevation, slope, aspect). There's no water signal to key off, since the channels are dry most of the year. 2. Validating the free satellite data. Airborne lidar is the accuracy benchmark but only exists for 2001 and 2014. Extending past 2014 means using REMA (free satellite elevation models). She showed REMA is good enough via ICP (a 3D alignment method) and NMAD (a median-based, outlier-robust spread estimate that's far more reliable here than standard deviation). 3. Measuring change. Differencing an old elevation surface from a newer one inside the channels gives a DEM-of-Difference (DoD). Only changes above the noise floor (LOD95 = 1.96 × NMAD) count. Summed per stream, that yields erosion/deposition rates and even acceleration.

The gap. Barlow's work ends at *description*. It maps where change happens (Denton Hills erodes hardest; parts of Taylor Valley are accelerating) but explicitly leaves two things for future work: (a) linking the change to its physical and climate drivers, and (b) making the model work beyond the MDVs. That's the natural space for a Future Investigator project: moving from a tool that *watches* to one that *explains and predicts*.

2. What we need to do

In one sentence: measure how fast each Antarctic stream is reshaping its channel, then explain that rate with heat and melt, so the map stops just *describing* change and starts *predicting* it. That breaks into three concrete objectives:

#	Objective	Hypothesis	Why it's new
O1	Attribution (the headline). Relate each stream's change rate to its drivers: temperature, insolation, positive-degree-days (PDD) (cumulative above-freezing heat), melt-season discharge, glacier melt, and active-layer (thaw) depth, drawn from the LTER field network plus ERA5/AMPS reanalysis.	H1: Change is governed by *cumulative melt energy*, not raw water volume. An energy-based model outperforms a discharge-only model and explains the scatter discharge leaves behind.	Barlow lists this as future work; it hasn't been quantified.

#	Objective	Hypothesis	Why it's new
O2	Generalization. Make the terrain-only detector work across both lidar (2001/2014) and REMA, and across all four valleys. The obstacle is domain shift: lidar and satellite represent the surface differently, which degrades the satellite epoch.	H2: A sensor-aware correction (conditioned on slope/aspect, per pixel) closes most of the lidar-vs-REMA gap.	Barlow flags sensor generalization as future work.
O3	Calibrated uncertainty. Replace the single global noise value with a per-pixel, sensor-aware one, so every change map carries a defensible confidence layer.	H3: Conditioning NMAD on slope, aspect, and sensor produces a 95% bound that actually holds 95% of the time.	Barlow uses one global value per epoch pair.

The unifying payoff: a model that takes a driver field and predicts where acceleration migrates next, which is the operationally useful form of the climate-indicator idea.

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3. The data: what we'll use, where it's from, what it shows

Everything below is free and public (the one exception is flagged). The datasets fall into three jobs: snapshots of the ground (to measure change), drivers (to explain it), and labels (to know where the streams are).

Three snapshots of the ground. Change detection needs the same terrain measured at different times. We have three epochs:

Dataset	Where it's from	What it shows	What it does for us
2001 airborne lidar (2 m)	NASA Airborne Topographic Mapper survey (via USGS)	The valley floors as they were in 2001	The baseline: every change is measured against this starting surface
2014 airborne lidar (1 m)	NCALM flight campaign (via OpenTopography)	The same terrain 13 years later, at the highest accuracy available	The benchmark epoch: sharpest data, anchors the noise model, midpoint of the record
REMA satellite DEM (2 m, 2021–23)	Polar Geospatial Center, built from Maxar stereo imagery	The most recent surface, covering all of Antarctica	Extends the record past the last lidar flight; since no new lidar is planned, this is the future of monitoring, which is why O2 (making it interchangeable with lidar) matters

From each snapshot we compute the terrain layers the detector reads: slope, aspect, profile curvature (signed, so concave channels separate from convex ridges), MFD flow accumulation (where water would collect), and lidar intensity.

Drivers: the "why" variables. O1 tests whether melt energy explains the measured change, so we need the energy and water records:

Dataset	Where it's from	What it shows	What it does for us
Stream gauges (21 streams, daily, 1990s–present)	McMurdo LTER field network (via EDI)	How much meltwater each stream actually carried, day by day	The direct water driver for O1
Meteorology stations	McMurdo LTER (via EDI)	Air temperature, solar radiation, wind on the valley floors	Builds the melt-energy drivers: positive-degree-days and insolation

Dataset	Where it's from	What it shows	What it does for us
Glacier mass balance (7 glaciers)	McMurdo LTER (via EDI)	How much ice each glacier gained or lost per season	Ties each stream's water supply to its source glacier
Soil temperature and moisture	McMurdo LTER (via EDI)	How deep the ground thaws each summer	The active-layer driver: thawed banks erode more easily
ERA5 reanalysis	ECMWF Copernicus Climate Data Store	Continuous modeled weather for every point, 1993–present	Fills the gaps between stations, and between gauged and ungauged streams
AMPS forecasts (2.67 km)	NCAR (via GDEX)	Antarctic-specific high-resolution atmosphere	Cross-checks ERA5 in the valleys' complicated terrain

Labels: where the streams are.

Dataset	Where it's from	What it shows	What it does for us
Stream channel polygons	McMurdo LTER GIS (EDI `knb-lter-mcm.6007`)	The mapped channel of every named stream	Masks the change maps to actual channels and stands in as training labels
Barlow's 217 hand-drawn tiles	Her research group (not public)	Expert-traced channel boundaries used to train her U-Net	The gold-standard training set, the one access-gated item; §6 covers the fallback
High-res satellite imagery (optional)	Polar Geospatial Center	Visual ground truth	Spot-checking labels and odd change patches

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4. How we'll do it

The data above feed three work packages, one per objective.

O1, attribution. For each gauged stream and epoch: (i) take the change rate from the DoD inside the channel; (ii) assemble a per-stream energy time series (PDD, insolation, discharge, thaw depth); (iii) fit rate-vs-driver models (hierarchical regression and random forest) with leave-one-stream-out validation; and (iv) test H1 directly: energy-based model against discharge-only. For streams with three time points, regress *acceleration* against *driver trends*.

O2, generalization. Quantify the lidar-vs-REMA difference over stable ground as a function of slope and aspect, learn a correction, retrain the U-Net with both sensors represented, and evaluate F1 across all four valleys and both sensors. Prior work (§5) shows this architecture holding up on an entirely different landscape, which bounds the risk.

O3, uncertainty. Replace the global NMAD with one conditioned on (slope, aspect, sensor), output it as a per-pixel layer, and verify calibration by confirming the 95% bound is exceeded about 5% of the time on stable ground.

Reproducibility. The pipeline is fully scripted: re-run from source and the outputs match, with robust statistics and CRS checks enforced throughout.

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5. FI qualifications

The FI has built and run this same class of machinery (U-Net segmentation on terrain layers, ICP alignment, DoD change detection) on a very different landscape: detecting channels, roads, and disturbance scars in the Appalachian Plateau from elevation alone. That shows the approach transfers across sensors and biomes, which is precisely O2's risk, and that the FI can operate the full pipeline end to end.

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6. Plan, timeline, risks, and data sharing

My role and development. I'm the lead and main author; my advisor mentors and is PI of record. I'll present at AGU and a SCAR/ISAES Antarctic venue, publish the O1 attribution result as its own paper, and take coursework in Bayesian/hierarchical modeling and remote-sensing uncertainty.

Timeline (3 years).

Year	Deliverables
Yr 1	Build the change-detection chain from public data; build the per-stream driver time series; first O1 attribution model on Taylor Valley; prototype the per-pixel uncertainty layer (O3). Output: attribution paper drafted.
Yr 2	Correct the lidar↔REMA domain shift and generalize the detector to all four valleys (O2); run attribution basin-wide; AGU talk. Output: O2 paper.
Yr 3	Predict where acceleration migrates next; full three-epoch acceleration attribution; release the uncertainty product (O3). Output: synthesis/prediction paper plus public datasets.

Risks and mitigations. (1) *Barlow's training labels are gated.* Request them from her group; if that stalls, the public LTER centerlines serve as a stand-in, and the FI's prior work shows label-building from scratch is in hand. (2) *REMA coverage after 2014 is uneven.* Prioritize Taylor Valley (full three-epoch coverage) for the acceleration work; treat other valleys as two-epoch. (3) *Gauge records have gaps.* That's a core reason O1 relies on melt *energy* from reanalysis rather than raw gauge discharge alone.

Data sharing. All inputs are free/public and cited. My outputs (change maps, per-stream rate tables, uncertainty layers, stream-boundary polygons) will be released open-access (DOIs via Zenodo/EDI) alongside the code. Heavy regenerable rasters stay out of version control; scripts, figures, and tables are tracked. Everything carries explicit CRS, method, and uncertainty metadata, consistent with NASA's open-science requirements.

References

[1] Barlow, M. C. (2026). *A Comprehensive Spatial Analysis of Stream Boundary and Geomorphological Change Detection: McMurdo Dry Valleys, Antarctica.* PhD Dissertation, Univ. of Houston (chair: C. L. Glennie). 240 pp.

[2] Barlow, M. C., Zhu, X., & Glennie, C. L. (2022). Stream Boundary Detection of a Hyper-Arid, Polar Region Using a U-Net Architecture: Taylor Valley, Antarctica. *Remote Sensing* 14(1):234. doi:10.3390/rs14010234. *(Dissertation Ch. 4, the proof of concept.)*

[3] Höhle, J., & Höhle, M. (2009). Accuracy assessment of DEMs by means of robust statistical methods. *ISPRS J. Photogramm. Remote Sens.* 64(4):398–406. *(Origin of NMAD.)*

[4] Howat, I. M., et al. (2019). The Reference Elevation Model of Antarctica (REMA). *The Cryosphere* 13:665–674.

[5] Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. *MICCAI* 234–241.

[6] McMurdo Dry Valleys LTER datasets (stream discharge, meteorology, glacier mass balance, soil), Environmental Data Initiative, `knb-lter-mcm.*`.

[7] Hersbach, H., et al. (2020). The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.* 146:1999–2049.

Plain-language version of `finesst_proposal.md`*: same plan, more direct voice. Written as a plan only (no preliminary work presented). Not a submitted proposal.*